

TP 5-II-Pratice Exercise(ClassicModels)

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Part A:SQL Basics

Simple SELECT

Only from USA

```
SELECT customerName, country, creditLimit
FROM customers
WHERE country = 'USA';
```

Result:

```
| customerName          | country | creditLimit |
+-----+-----+-----+
| Signal Gift Stores    | USA     | 71800       |
| Mini Gifts Distributors Ltd. | USA     | 210500      |
| Mini Wheels Co.       | USA     | 64600       |
| Land of Toys Inc.     | USA     | 114900      |
| Muscle Machine Inc    | USA     | 138500      |
| Diecast Classics Inc. | USA     | 100600      |
| Technics Stores Inc.  | USA     | 84600       |
| American Souvenirs Inc | USA     | 0           |
```

ORDER BY

Order by buyPrice DESC

```
SELECT productName, buyPrice, MSRP
FROM products
ORDER BY buyPrice DESC;
```

Result:

productName	buyPrice	MSRP
1962 LanciaA Delta 16V	103.42	147.74
1998 Chrysler Plymouth Prowler	101.51	163.73
1952 Alpine Renault 1300	98.58	214.3
1956 Porsche 356A Coupe	98.3	140.43
2001 Ferrari Enzo	95.59	207.8
1968 Ford Mustang	95.34	194.57
1995 Honda Civic	93.89	142.25
1970 Triumph Spitfire	91.92	143.62
2003 Harley-Davidson Eagle Drag Bike	91.02	193.66
1969 Corvair Monza	89.14	151.08
1917 Grand Touring Sedan	86.7	170

WHERE + BETWEEN

amount is between 1000 and 50000 and sort by paymentDate

```
SELECT *
FROM payments
WHERE amount BETWEEN 10000 AND 50000
ORDER BY paymentDate;
```

Result:

customerNumber	checkNumber	paymentDate	amount
363	IS232033	2003-01-16	10223.83
128	DI925118	2003-01-28	10549.01
141	JN722010	2003-02-25	40206.2
131	CL442705	2003-03-12	22292.62
486	JB117768	2003-03-20	25833.14
187	KL124726	2003-03-27	48425.69
129	PI42991	2003-04-09	16537.85
124	CQ287967	2003-04-11	11044.3
424	LM271923	2003-04-16	21665.98
172	EH208589	2003-04-20	33383.14

Part B:JOIN Practice

Ex4:Customer & Orders

**display:customerName, ,orderNumber ,orderDate and status **

```
SELECT customers.customerName, orders.orderNumber, orders.orderDate,  
orders.status  
-> FROM customers  
-> JOIN orders ON customers.customerNumber = orders.customerNumber;
```

Result:

customerName	orderNumber	orderDate	status
Online Diecast Creations Co.	10100	2003-01-06	Shipped
Blauer See Auto, Co.	10101	2003-01-09	Shipped
Vitachrome Inc.	10102	2003-01-10	Shipped
Baane Mini Imports	10103	2003-01-29	Shipped
Mini Auto Werke	10164	2003-10-21	Resolved
Dragon Souveniers, Ltd.	10165	2003-10-22	Shipped
FunGiftIdeas.com	10166	2003-10-21	Shipped
Scandinavian Gift Ideas	10167	2003-10-23	Cancelled

ex5:Orders & Product

Show: orderNumber, productName, quantityOrdered and priceEach

```
SELECT  
    od.orderNumber,  
    p.productName,  
    od.quantityOrdered,  
    od.priceEach  
FROM orders o  
JOIN orderdetails od ON o.orderNumber = od.orderNumber  
JOIN products p ON od.productCode = p.productCode;
```

Result:

	orderNumber	productName	
	quantityOrdered	priceEach	
	10123	1965 Aston Martin DB5	
26	120.71		
	10123	1999 Indy 500 Monte Carlo SS	
46	114.84		
	10123	1948 Porsche Type 356 Roadster	
34	117.26		
	10123	1966 Shelby Cobra 427 S/C	
50	43.27		
	10298	1996 Moto Guzzi 1100i	
39	105.86		
	10298	1936 Harley Davidson El Knucklehead	
32	60.57		
	10345	1938 Cadillac V-16 Presidential Limousine	
43	38.98		

Ex6:Employee & Office

Display: employee full name, jobTitle, oce city and oce country

```
SELECT
    CONCAT(e.firstName, ' ', e.lastName) AS 'Employee Full Name',
    e.jobTitle,
    o.city,
    o.country
FROM employees e
JOIN offices o ON e.officeCode = o.officeCode;
```

Result:

Employee Full Name	jobTitle	city	country
Diane Murphy	President	San Francisco	USA
Mary Patterson	VP Sales	San Francisco	USA
Jeff Firrelli	VP Marketing	San Francisco	USA
William Patterson	Sales Manager (APAC)	Sydney	Australia
Gerard Bondur	Sale Manager (EMEA)	Paris	France
Anthony Bow	Sales Manager (NA)	San Francisco	USA

Part C:Function(Student Creates Function)

ex7:Function-Customer Credit Lever

Create a function `creditLevel(creditLimit DOUBLE)` Rules: `creditLimit < 50,000` → Low, `50,000–100,000` → Medium and `100,000` → High

```
DELIMITER $$

CREATE FUNCTION creditLevel(creditLimit DOUBLE)
RETURNS VARCHAR(10)
DETERMINISTIC
BEGIN
    DECLARE lvl VARCHAR(10);

    IF creditLimit < 50000 THEN
        SET lvl = 'Low';
    ELSEIF (creditLimit >= 50000 AND creditLimit <= 100000) THEN
        SET lvl = 'Medium';
    ELSEIF creditLimit > 100000 THEN
        SET lvl = 'High';
    END IF;

    RETURN (lvl);
END$$

DELIMITER ;
```

Call function

```
SELECT customerName, creditLevel(creditLimit)
FROM customers;
```

Result:

customerName	creditLevel(creditLimit)
Atelier graphique	Low
Signal Gift Stores	Medium
Australian Collectors, Co.	High
La Rochelle Gifts	High
Baane Mini Imports	Medium
Mini Gifts Distributors Ltd.	High
Havel & Zbyszek Co	Low
Blauer See Auto, Co.	Medium

Ex8:Function-Total Order Amount

Create a function `totalOrderAmount(orderNo INT)` Return the **total price** of one order

```
--Create function
DELIMITER $$

CREATE FUNCTION totalOrderAmount(orderNo INT)
RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
    DECLARE total DECIMAL(10,2);

    SELECT SUM(quantityOrdered * priceEach)
    INTO total
    FROM orderdetails
    WHERE orderNumber = orderNo;

    RETURN IFNULL(total, 0);
END$$

DELIMITER ;

--Call funcion

SELECT
    orderNumber,
    totalOrderAmount(orderNumber) AS 'Total Amount'
FROM orders;
```

Result:

orderNumber	Total Amount
10123	14571.44
10298	6066.78
10345	1676.14
10124	32641.98
10278	33347.88
10346	14191.12
10120	45864.03

Part D:Store Procedures

Ex9:Customers by Country

```
--Create procedure

DELIMITER $$

CREATE PROCEDURE getCustomersByCountry(IN countryName VARCHAR(50))
```

```

BEGIN
    SELECT customerName, phone, city
    FROM customers
    WHERE country = countryName;
END$$

DELIMITER ;

--Call
CALL getCustomersByCountry('France');

```

Return: *customerName, phone and city*

Result:

customerName	phone	city
Atelier graphique	40.32.2555	Nantes
La Rochelle Gifts	40.67.8555	Nantes
Saveley & Henriot, Co.	78.32.5555	Lyon
Daedalus Designs Imports	20.16.1555	Lille
La Corne D'abondance, Co.	(1) 42.34.2555	Paris
Mini Caravy	88.60.1555	Strasbourg
Alpha Cognac	61.77.6555	Toulouse
Lyon Souvenirs	+33 1 46 62 7555	Paris
Auto Associés & Cie.	30.59.8555	Versailles
Marseille Mini Autos	91.24.4555	Marseille
Reims Collectables	26.47.1555	Reims
Auto Canal+ Petit	(1) 47.55.6555	Paris

Ex10:Product by Price Range

Return: *productName, buyPrice and MSRP Sorted by buyPrice ASC.*

```

--Create
DELIMITER $$

CREATE PROCEDURE getProductsByPrice(
    IN minPrice DECIMAL(10,2),
    IN maxPrice DECIMAL(10,2)
)
BEGIN
    SELECT productName, buyPrice, MSRP
    FROM products
    WHERE buyPrice BETWEEN minPrice AND maxPrice
    ORDER BY buyPrice ASC;

```

```
END$$

DELIMITER ;
--Call
CALL getProductsByPrice(50, 100);
```

Result:

productName	buyPrice	MSRP
1969 Chevrolet Camaro Z28	50.51	85.61
The Titanic	51.09	100.17
American Airlines: B767-300	51.15	91.34
1999 Yamaha Speed Boat	51.61	86.02
1904 Buick Runabout	52.66	87.77
The Queen Mary	53.63	99.31
1948 Porsche 356-A Roadster	53.9	77
1980's GM Manhattan Express	53.93	96.31
F/A 18 Hornet 1/72	54.4	80

Ex11:Order by Status

Return: orderNumber, orderDate and customerName

```
--Create
DELIMITER $$

CREATE PROCEDURE getOrdersByStatus(IN orderStatus VARCHAR(15))
BEGIN
    SELECT
        o.orderNumber,
        o.orderDate,
        c.customerName
    FROM orders o
    JOIN customers c ON o.customerNumber = c.customerNumber
    WHERE o.status = orderStatus;
END$$

DELIMITER ;
--Call
CALL getOrdersByStatus('Shipped');
```

Result:

orderNumber	orderDate	customerName
10100	2003-01-06	Online Diecast Creations Co.
10101	2003-01-09	Blauer See Auto, Co.
10102	2003-01-10	Vitachrome Inc.
10103	2003-01-29	Baane Mini Imports
10104	2003-01-31	Euro+ Shopping Channel
10105	2003-02-11	Danish Wholesale Imports
10106	2003-02-17	Rovelli Gifts
10107	2003-02-24	Land of Toys Inc.
10108	2003-03-03	Cruz & Sons Co.
10109	2003-03-10	Motor Mint Distributors Inc.
10110	2003-03-18	AV Stores, Co.

PartE: Logic & Loop Practice

Ex12:Prime Number Function

Create function *isPrime(num INT)* Return: "X is prime" or "X is not prime"

```
--Create function
DELIMITER $$

CREATE FUNCTION isPrime(num INT)
RETURNS VARCHAR(20)
DETERMINISTIC
BEGIN
    DECLARE i INT DEFAULT 2;
    IF num <= 1 THEN RETURN CONCAT(num, ' is not prime'); END IF;

    WHILE i <= SQRT(num) DO
        IF num % i = 0 THEN
            RETURN CONCAT(num, ' is not prime');
        END IF;
        SET i = i + 1;
    END WHILE;

    RETURN CONCAT(num, ' is prime');
END$$

DELIMITER ;

--Test
Select isPrime(11);
```

```
| isPrime(11) |
+-----+
```

```
| 11 is prime |
+-----+
```

Ex13:Count Customer Orders

Create function `countOrders(customerNo INT)` Return number of orders per customer.

```
--Create
DELIMITER $$

CREATE FUNCTION countOrders(customerNo INT)
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE orderCount INT;

    SELECT COUNT(*)
    INTO orderCount
    FROM orders
    WHERE customerNumber = customerNo;

    RETURN IFNULL(orderCount, 0);
END$$

DELIMITER ;

--Test
SELECT
    customerName,
    countOrders(customerNumber) AS total_orders
FROM customers;
```

Result:

customerName	total_orders
Atelier graphique	3
Signal Gift Stores	3
Australian Collectors, Co.	5
La Rochelle Gifts	4
Baane Mini Imports	4
Mini Gifts Distributors Ltd.	17
Havel & Zbyszek Co	0
Blauer See Auto, Co.	4

Part F:Advance(Optional/Bonus)

Ex14:TOP Customers

Return *top 5 customers* by total payment amount

```

---Create
DELIMITER $$

CREATE PROCEDURE topCustomers(IN topLimit INT)
BEGIN
    SELECT
        c.customerName,
        SUM(p.amount) AS totalPayment
    FROM customers c
    JOIN payments p ON c.customerNumber = p.customerNumber
    GROUP BY c.customerNumber
    ORDER BY totalPayment DESC
    LIMIT topLimit;
END$$

DELIMITER ;
--call
Call topCustomers(5);

```

Result:

customerName	totalPayment
Euro+ Shopping Channel	715738.9800000001
Mini Gifts Distributors Ltd.	584188.2400000001
Australian Collectors, Co.	180585.06999999998
Muscle Machine Inc	177913.95
Dragon Souvenirs, Ltd.	156251.03

Ex15:Drop Table Procedure

Uses *dynamic SQL* safely.

```

# Tables
| Tables_in_classicmodels |
+-----+
| customers                |
| employees                |
| offices                  |
| orderdetails             |
| orders                   |
| users                    |
| payments                 |

```

```
| productlines      |
| products          |
+-----+
```

```
--Create
DELIMITER $$

CREATE PROCEDURE dropTableName(IN tName VARCHAR(64))
BEGIN
    SET @s = CONCAT('DROP TABLE IF EXISTS ', tName);
    PREPARE stmt FROM @s;
    EXECUTE stmt;
    DEALLOCATE PREPARE stmt;
END$$

DELIMITER ;
--Call
CALL dropTableName('users');
```

Result:

```
| Tables_in_classicmodels |
+-----+
| customers                |
| employees                |
| offices                  |
| orderdetails             |
| orders                   |
| payments                 |
| productlines             |
| products                 |
+-----+
```

The End , THANK YOU!!
